

Secure-Tunnel: Post-Quantum Cryptographic Overlay for UAV C2 Links

Results and Analysis Section — v4.0 (Cross-Axis Interactions)

Draft compilation for review

TABLE I: Algorithm Aliases and NIST Security Levels

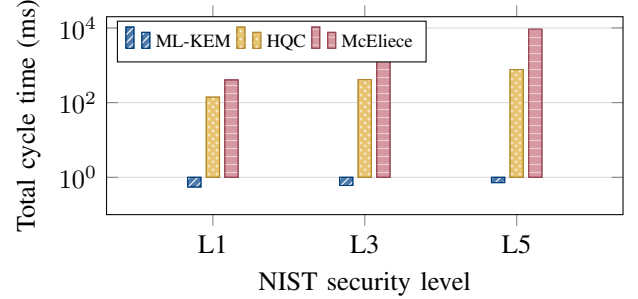
Alias	Full Name	Level	Type
ML512	ML-KEM-512	L1	KEM
ML768	ML-KEM-768	L3	KEM
ML1024	ML-KEM-1024	L5	KEM
HQ128	HQC-128	L1	KEM
HQ192	HQC-192	L3	KEM
HQ256	HQC-256	L5	KEM
MC348	Classic-McEliece-348864	L1	KEM
MC460	Classic-McEliece-460896	L3	KEM
MC8192	Classic-McEliece-8192128	L5	KEM
DSA44	ML-DSA-44	L1	SIG
DSA65	ML-DSA-65	L3	SIG
DSA87	ML-DSA-87	L5	SIG
F512	Falcon-512	L1	SIG
F1024	Falcon-1024	L5	SIG
SP128	SPHINCS ⁺ -SHA2-128s	L1	SIG
SP192	SPHINCS ⁺ -SHA2-192s	L3	SIG
SP256	SPHINCS ⁺ -SHA2-256s	L5	SIG
AESG	AES-256-GCM	—	AEAD
CH20	ChaCha20-Poly1305	—	AEAD
ASC	Ascon-128a	—	AEAD

I. RESULTS AND ANALYSIS

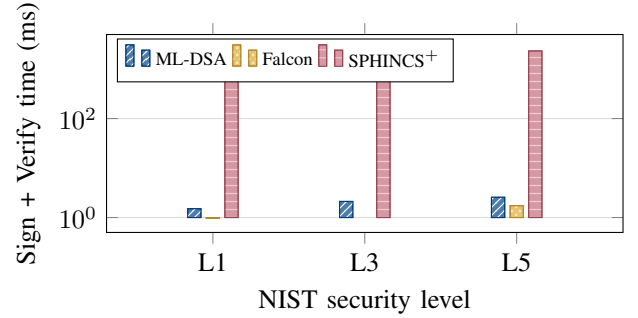
All measurements were collected on a Raspberry Pi 4 Model B Rev 1.5 (BCM2711, four Cortex-A72 cores at 1.8 GHz, 4 GB RAM) running Debian 12 (kernel 6.12.47) with the performance CPU governor. Timing used `time.perf_counter_ns()` with garbage collection disabled; power was sampled via an INA219 current sensor (0.1 Ω shunt, V_{BUS} gain correction 1.22) at up to 1 kHz. Idle power: $P_{idle} = 3.32$ W (30 s baseline). The Cortex-A72 lacks ARMv8 Crypto Extensions; AES-GCM therefore executes in software table-lookup mode, while ChaCha20-Poly1305 exploits 128-bit NEON SIMD. PQC primitives: liboqs-python 0.14.0. AEAD ciphers: OpenSSL 3.5.4 (AES-256-GCM, ChaCha20-Poly1305), ascon-c v1.2 opt64 via ctypes (Ascon-128a, KAT-verified). Table I defines the short aliases used throughout.

A. Primitive-Level Performance

Key Encapsulation. Fig. 1a and Table II compare nine KEM algorithms across three NIST security levels (100 iterations, 1 kHz INA219). Each cycle comprises keygen, encaps-



(a) KEM: keygen + encaps + decaps



(b) SIG: sign + verify (Falcon unavailable at L3)

Fig. 1: Primitive-level computational cost by NIST level (log scale). ML-KEM and ML-DSA/Falcon complete in < 3 ms at all levels; SPHINCS⁺ and McEliece are 2–4 orders of magnitude slower.

sulation, and decapsulation—all three are executed per rekey event. Fig. 1 presents both KEM and signature results using consistent log-scale grouping.

ML-KEM completes a full cycle under 0.72 ms through Level 5, drawing 3.46–3.48 W. HQC is 255–1 066 $\times$ slower, driven by decapsulation (73–393 ms). Classic McEliece exhibits extreme keygen asymmetry: MC8192 requires 9.02 s for keygen but only 1.89 ms for encapsulation, making it feasible only with pre-provisioned keys.

Digital Signatures. Fig. 1b and Table III evaluate eight signature schemes. Keygen is a one-time cost amortized over the key lifetime (0.4–282 ms); the per-handshake sign and

TABLE II: KEM Per-Operation Breakdown (100 iter, 1 kHz INA219)

Alias	Lvl	KeyGen (ms)	Encaps (ms)	Decaps (ms)	E_{tot} (μJ)	ρ
ML512	L1	0.240	0.152	0.158	1907	$1.00\times$
HQ128	L1	22.33	44.68	73.39	535 253	$255\times$
MC348	L1	351.0	0.466	55.98	1 737 186	$741\times$
ML768	L3	0.237	0.182	0.185	2 112	$1.00\times$
HQ192	L3	67.68	135.8	211.9	1 651 813	$688\times$
MC460	L3	1 281	0.872	89.59	6 025 434	$2 272\times$
ML1024	L5	0.273	0.216	0.231	2 507	$1.00\times$
HQ256	L5	124.4	250.2	392.6	3 136 867	$1 066\times$
MC8192	L5	9 021	1.894	210.1	40 583 491	$12 823\times$

E_{tot} : total energy (keygen+encaps+decaps), INA219-measured. ρ : total cycle time relative to ML-KEM at same level.

TABLE III: Signature Runtime Cost: Sign + Verify (100 iter, 1 kHz INA219). KeyGen is a one-time cost and excluded from ρ .

Alias	Lvl	Sign (ms)	Verify (ms)	E_s (μJ)	E_v (μJ)	ρ
DSA44	L1	1.179	0.331	4 264	1 175	$1.00\times$
F512	L1	0.778	0.192	2 773	683	$0.64\times$
SP128	L1	1 468	1.582	6 008 719	5 690	$972\times$
DSA65	L3	1.633	0.484	5 849	1 723	$1.00\times$
SP192	L3	2 610	2.299	10 628 558	8 318	$1 234\times$
DSA87	L5	1.847	0.723	6 612	2 564	$1.00\times$
F1024	L5	1.452	0.280	5 175	995	$0.67\times$
SP256	L5	2 321	3.292	9 516 164	11 738	$904\times$

ρ : combined sign+verify time relative to ML-DSA at same level. Falcon $\rho < 1$: faster than ML-DSA. E_s/E_v : INA219-measured energy per operation.

verify operations determine rekey latency.

Falcon achieves the lowest combined sign+verify latency at L1 (0.970 ms) and L5 (1.732 ms), outperforming ML-DSA by 36% and 33% respectively. SPHINCS⁺ imposes prohibitive signing latency: SP128 consumes 1.47 s and 6.01 J per signature, approaching the rekey period at short intervals. Verification is fast (≤ 3.3 ms) for all algorithms, confirming the ground-station path is never the bottleneck.

Takeaway: Only ML-KEM paired with ML-DSA or Falcon enables sub-millisecond primitive costs at all NIST levels on resource-constrained ARM hardware.

B. Protocol-Level Handshake Performance

Each of the 24 level-matched KEM $\times$ SIG combinations was benchmarked end-to-end on the tunnel prototype (10 s per suite, baseline with no DDoS detector). T_{HS} is the measured wall-clock time from handshake initiation to session establishment, including protocol framing, HKDF key derivation, and interpreter overhead. T_{crypto} is the sum of in-suite KEM and SIG primitive times recorded during the same run.

Fig. 2 compares best-case handshake time per KEM family at each security level. Table IV reports all 24 suites.

The default suite (ML768+DSA65) completes a full handshake in 12.71 ms at Level 3, of which only 2.71 ms (21%) is cryptographic computation; the remaining ~ 10 ms comprises

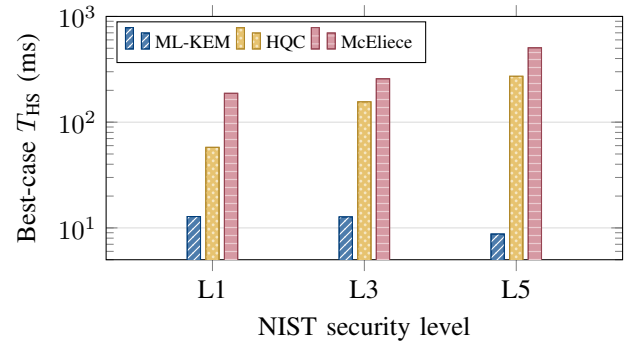


Fig. 2: Best-case measured handshake time per KEM family and NIST level (fastest SIG pairing, log scale). ML-KEM achieves < 13 ms at all levels; McEliece exceeds 500 ms at L5.

protocol framing, HKDF derivation, and interpreter overhead. ML1024+F1024 at Level 5 measures 8.75 ms—the fastest of all 24 suites—because Falcon-1024’s combined sign+verify (1.18 ms) undercuts ML-DSA-65 (1.94 ms).

Three structural findings emerge: (1) ML-KEM suites are the only family achieving sub-100 ms handshakes at every level ($\rho \leq 1.9\times$); (2) SPHINCS⁺ renders any suite non-viable regardless of KEM choice ($T_{\text{HS}} \geq 655$ ms); (3) McEliece at L5 exceeds 500 ms due to keygen asymmetry. The MC460+SP192 timeout at 48.2 s empirically justifies the scheduler’s constraint forbidding McEliece $\times$ SPHINCS⁺ pairs.

Takeaway: Protocol overhead dominates crypto cost for lattice-based suites; ML-KEM $\times$ ML-DSA/Falcon handshakes complete under 17 ms through NIST Level 5.

C. Data-Plane AEAD Performance

AEAD selection operates on the UDP data plane and does not affect handshake timing. Table V reports results from 10 000 iterations (500 warmup, two-pass methodology: timing with GC disabled, power with INA219 at ~ 100 Hz).

Timing is concentrated (σ_{enc} : AESG 0.38 μs , ASC 0.77 μs). CH20 exhibits $\sigma = 3.99 \mu\text{s}$ from rare scheduling jitter (P95=6.46, P99=6.69 μs ; median 6.30 μs).

Fig. 3 shows latency scaling from MAVLink telemetry (9 B) to bulk transfer (4096 B).

At 4096 B, CH20 is $3.93\times$ faster than AESG due to NEON vectorization versus software AES table lookups. Below 64 B, setup cost dominates and the ciphers converge. Energy per bit at 256 B: CH20=21.8, AESG=32.2, ASC=46.4 nJ/bit. For a 50 Hz MAVLink stream (~ 256 B packets), per-second AEAD cost ranges from 2.23 mJ (CH20) to 4.75 mJ (ASC)—negligible versus the 3.32 W board draw.

Takeaway: ChaCha20-Poly1305 delivers 32% lower energy per bit than AES-256-GCM on Cortex-A72 without hardware AES, making it the optimal data-plane cipher for this platform.

D. Adversarial Interaction Impact

Concurrent DDoS detection competes with the cryptographic pipeline for CPU and thermal headroom. All 72

TABLE IV: Measured End-to-End Suite Handshake Performance — All 24 Level-Matched KEM×SIG Combinations (Baseline, No DDoS Detector)

KEM	SIG	T_{HS} (ms)	T_{crypto} (ms)	$\hat{E}_{\text{HS}}$ (mJ)	Crypto (%)	ρ	Viable?
<i>NIST Level 1</i> (3 KEMs × 3 SIGs = 9 suites)							
ML512	F512	12.80	5.69	44.3	44	1.0×	✓
ML512	DSA44	14.34	2.45	49.6	17	1.1×	✓
HQ128	F512	57.80	54.30	200	94	4.5×	✓
HQ128	DSA44	61.43	56.50	213	92	4.8×	✓
MC348	DSA44	187.5	132.3	649	71	14.6×	✓*
MC348	F512	206.1	161.4	713	78	16.1×	✓*
ML512	SP128	655.1	644.8	2 268	98	51.2×	×
HQ128	SP128	722.8	718.1	2 502	99	56.5×	×
MC348	SP128	1 080	1 045	3 739	97	84.3×	×
<i>NIST Level 3</i> (3 KEMs × 2 SIGs = 6 suites)							
ML768	DSA65	12.71	2.71	44.3	21	1.0×	✓
HQ192	DSA65	155.8	159.9	543	100 [‡]	12.3×	✓*
MC460	DSA65	257.4	185.4	897	72	20.2×	✓*
ML768	SP192	1 250	1 233	4 362	99	98.3×	×
HQ192	SP192	1 537	1 533	5 364	100	120.9×	×
MC460	SP192	48 186 [†]	—	—	—	—	×
<i>NIST Level 5</i> (3 KEMs × 3 SIGs = 9 suites)							
ML1024	F1024	8.75	1.82	30.5	21	1.0×	✓
ML1024	DSA87	16.78	2.25	58.5	13	1.9×	✓
HQ256	F1024	271.7	283.7	947	100 [‡]	31.1×	✓*
HQ256	DSA87	276.1	288.9	963	100 [‡]	31.6×	✓*
MC8192	DSA87	505.3	367.3	1 763	73	57.7×	×
MC8192	F1024	678.4	527.9	2 367	78	77.5×	×
ML1024	SP256	1 110	1 092	3 873	98	126.9×	×
HQ256	SP256	1 470	1 494	5 128	100 [‡]	168.0×	×
MC8192	SP256	1 651	1 525	5 759	92	188.7×	×

T_{HS} : measured end-to-end handshake time. T_{crypto} : sum of in-suite KEM and SIG primitive times. $\hat{E}_{\text{HS}}$: estimated energy ($P_{\text{avg}} \times T_{\text{HS}}$, using per-primitive power from Tables II–III). ρ : relative to fastest suite at same NIST level. Viable: ✓ $T_{\text{HS}} < 100$ ms; ✓* extended rekey; × $T_{\text{HS}} > 500$ ms or timeout. [†]Timeout at 48.2 s; crypto fields unavailable. [‡] $T_{\text{crypto}} > T_{\text{HS}}$ (measurement jitter); capped at 100%.

TABLE V: AEAD Performance at 256-Byte Payload (10 000 iter, RPi4). Independent verification run; native C Ascon-128a backend (ascon-c v1.2 opt64).

Cipher	Enc (μs)	Dec (μs)	P_{enc} (W)	P_{dec} (W)	E_{pkt} (μJ)	E_{bit} (nJ/bit)
AESG	9.25	9.09	3.63	3.57	66.0	32.2
CH20	6.47	6.05	3.54	3.58	44.6	21.8
ASC	12.72	12.85	3.84	3.59	95.0	46.4

$E_{\text{pkt}} = P_{\text{avg}} \times (t_{\text{enc}} + t_{\text{dec}})$. $E_{\text{bit}} = E_{\text{pkt}}/2048$. Board-level power ($P_{\text{idle}} = 3.320$ W not subtracted). ΔP above idle: AESG 0.31, CH20 0.22, ASC 0.52 W.

level-matched suites were benchmarked under baseline (no detector), XGBoost, and TST scenarios (10 s per suite per scenario).¹

Table VI presents a head-to-head comparison of XGBoost and TST across accuracy, resource, and handshake-impact dimensions. Fig. 4 visualises the cross-axis cost interactions. Default-suite values are for ML768+DSA65+AESG.

Fig. 4 reveals a fundamental asymmetry in the three-axis interaction structure. Four cross-axis directions exist; their

¹XGBoost: 15 000 trees, 54 features, 15 CIC-IoT-2023 classes; TST: Time-Series Transformer, $\text{seq_len}=1$ (functionally a 3-layer MLP), 1.6 M parameters, 46 features, 6 coarse classes.

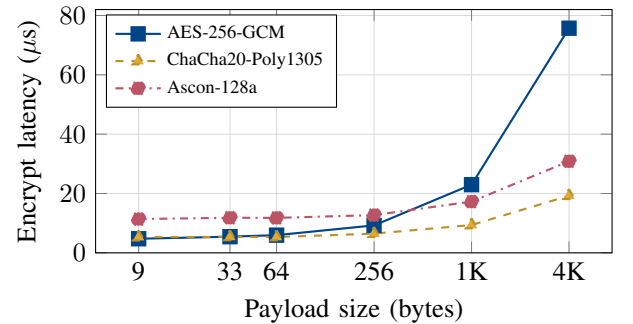


Fig. 3: AEAD encrypt latency versus payload size. ChaCha20 (NEON SIMD) scales at 3.4 $\mu\text{s}/\text{KiB}$; AES-GCM at 17.7 $\mu\text{s}/\text{KiB}$ (software table-lookup). Below 64 B, setup cost dominates and ciphers converge.

magnitudes differ by three orders of magnitude.

DDoS → PQC (handshake contention). Concurrent detection inflates the default-suite handshake: T_{HS} rises from 13.01 ms (baseline) to 13.46 ms (+3.5%) with XGBoost and 37.90 ms (+191%) with TST (Table VI). Across all 72 suites, XGBoost adds mean +2.5% / median +0.05% (max +137.2%); TST adds mean +71.8% / median +32.6% (max +405.1%). However, PQC handshakes occupy $< 0.05\%$ of each 30 s rekey

TABLE VI: XGBoost vs. TST Head-to-Head on RPi4 (CIC-IoT-2023)

Metric	XGBoost	TST	Ratio
<i>Detection quality</i>			
Accuracy (%)	94.55	90.27	—
F1-macro	0.947	0.80	—
<i>Inference cost (isolated, 15 s run)</i>			
t_{inf} mean (ms)	7.42	10.72	1.44×
t_{inf} P99 (ms)	9.52	26.12	2.74×
Throughput (pred/s)	134.6	93.2	0.69×
E_{inf} (mJ)	31.7	56.7	1.79×
<i>System impact (default suite)</i>			
CPU avg / peak (%)	40.2/61.4	87.8/94.1	2.2×/1.5×
ΔP above idle (mW)	+951	+1973	2.07×
ΔT (°C)	+14.1	+21.9	1.55×
T_{HS} (ms)	13.46	37.90	2.82×
HS overhead vs. baseline	+3.5%	+191%	55×

Idle baseline: $P_{\text{idle}} = 3.309$ W, CPU 14.7%, $T_{\text{HS}}^{\text{base}} = 13.01$ ms. CPU peak: max observed during 72-suite sweep. ΔT : rise over baseline for default suite (10 s run).

TABLE VII: AEAD Encrypt Latency Under Concurrent DDoS Detection (72-Suite Campaign Mean, RPi4)

Cipher	Baseline (μs)	+XGB (μs)	+TST (μs)	TST/Base
AESG	73.8	88.9 (+20%)	113.9 (+54%)	1.54×
CH20	71.2	86.8 (+22%)	105.3 (+48%)	1.48×
ASC	1353.7	1510.5 (+12%)	2165.9 (+60%)	1.60×

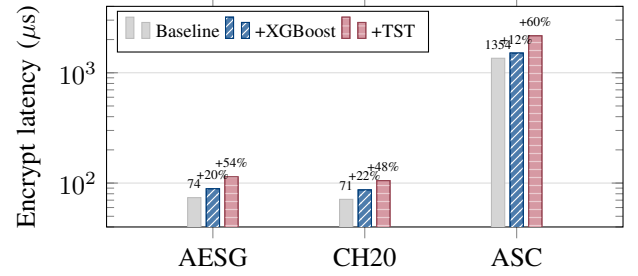
Values from `analysis_deep_dive.csv`. ASC measured with pure-Python backend as deployed during the DDoS campaign; native C backend (Table V) is 100× faster but was not used in this cross-axis campaign. Ascon is forbidden under both detectors by the MDEAS.

cycle ($\Phi < 5 \times 10^{-4}$, Eq. 1), so this penalty is transient.

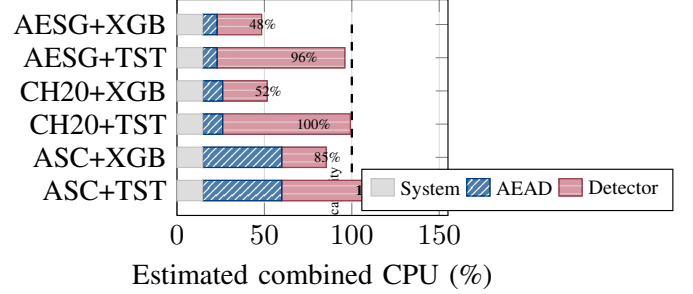
DDoS → AEAD (data-plane contention). This is the *dominant* cross-axis cost because both subsystems run continuously. Fig. 4a and Table VII report per-packet encrypt latency under concurrent detector load (72-suite campaign mean). XGBoost imposes 12–22% overhead across all three ciphers; TST imposes 48–60%. The log-scale grouping in Fig. 4a reveals that Ascon’s baseline latency (1354 μs) already exceeds AES-GCM’s worst-case under TST (114 μs) by 12×, compounding the cross-axis penalty.

AEAD → DDoS (CPU headroom). Fig. 4b decomposes the CPU budget into system, AEAD, and detector contributions. At 50 Hz MAVLink: AES-GCM uses 8.2%, ChaCha20 11.4%, and Ascon 45.2% CPU. XGBoost adds ~25.5% and TST ~73.1% above system idle. Combined, Ascon+TST would demand ~133% CPU—crossing the single-core capacity line (Fig. 4b, dashed). Even Ascon+XGBoost reaches 85%, leaving negligible thermal margin. The MDEAS therefore forbids Ascon under TST and flags it as marginal under XGBoost at rates > 200 Hz.

PQC → DDoS (rekey burst). During the 13 ms ML-KEM handshake, no inference runs; but $\Phi(30 \text{ s}) < 0.05\%$ (Eq. 1), so the detection gap is one missed sample at 100 Hz—negligible. McEliece keygen (up to 9 s) would block inference for an



(a) DDoS → AEAD: per-packet encrypt latency under concurrent detection (72-suite campaign mean, log scale). Labels show absolute μs (baseline) or overhead %. All three ciphers degrade; TST imposes 48–60% overhead.



(b) AEAD → DDoS: estimated CPU breakdown for each cipher×detector pair. System idle = 14.7%; AEAD and detector portions are additive. ASC+TST exceeds single-core capacity (dashed line)—forbidden by MDEAS.

Fig. 4: Bidirectional AEAD↔DDoS cross-axis interaction. (a) Detector CPU contention inflates per-packet AEAD latency by 12–22% (XGBoost) to 48–60% (TST). (b) Heavy ciphers consume the headroom needed by detectors; only AES-GCM and ChaCha20 leave sufficient budget for both detector types.

extended window, reinforcing the ML-KEM preference.

Takeaway: The dominant cross-axis penalty is DDoS↔AEAD because both run continuously. PQC interactions are transient ($\Phi < 0.05\%$) for ML-KEM suites. XGBoost is the only viable detector: it limits AEAD overhead to 20–22% at 4.3 pp higher accuracy than TST.

E. Scheduler Implications

The Multi-Dimensional Energy-Aware Scheduler (MDEAS) manages three decision axes at 1 Hz: AEAD cipher selection, security-level promotion/demotion (triggering PQC rekey), and DDoS detector activation. Each level transition requires a full handshake. The rekey overhead fraction is:

$$\Phi(R) = \frac{T_{\text{HS}}}{R + T_{\text{HS}}} \quad (1)$$

where R is the rekey interval. For the default suite ($T_{\text{HS}} = 12.71$ ms) at the minimum 30 s interval: $\Phi = 4.2 \times 10^{-4}$ (<0.05%). Even ML768+SP192 ($T_{\text{HS}} = 1.25$ s) at 30 s yields $\Phi = 0.040$ (4.0%).

Table VIII maps each detector state to the thermal activation ceiling used by the MDEAS.

TABLE VIII: MDEAS Detector Thermal Activation Ceilings

Detector	ΔP (W)	ΔT (°C)	CPU peak (%)	$T_{\max}$ (°C)
None	0.00	0.0	36.5	80.0
XGBoost	0.95	14.1	61.4	65.9
TST	1.97	21.9	94.1	58.1

$T_{\max} = 80^{\circ}\text{C} - \Delta T$: temperature ceiling before activation is blocked. ΔT , CPU peak from default-suite measurements (Table VI).

Cross-axis constraints prevent pathological configurations: Ascon-128a is forbidden under TST (per-packet cost plus 93% CPU exceeds the real-time budget); McEliece $\times$ SPHINCS⁺ pairs are forbidden (T_{HS} exceeds the 45 s rekey timeout); detector activation is blocked above the per-detector $T_{\max}$. These constraints reduce 648 theoretical configurations (216 suites $\times$ 3 detector states) to $\sim$ 500 viable operating points.

Takeaway: Rekey overhead is negligible ($< 0.05\%$) for ML-KEM suites. The MDEAS navigates the security–energy–thermal Pareto surface via 1 Hz telemetry while guaranteeing the configured minimum security level.